IP marshrutlash

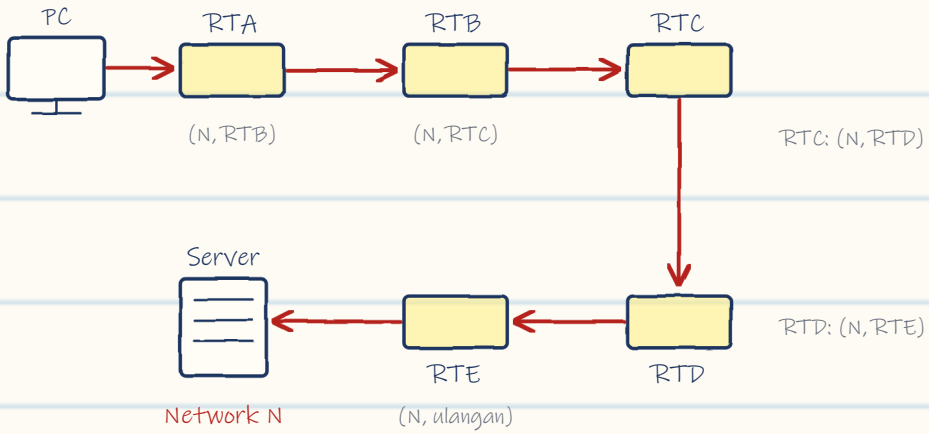
IP Routing

Asosiy atamalar

- Marshrut — paketni manzilga olib boradigan yo'l ma'lumoti
- Marshrut jadvali — routerdagi barcha marshrutlar ro'yxati
- Next hop — paket beriladigan keyingi qurilma manzili
- Metrika — yo'lning xarajati; kichigi afzal
- Preference — protokolga ishonch darajasi; kichigi afzal
- Longest match — eng uzun mos maska bo'yicha tanlash

1. Marshrutlash nima?

- Marshrut — paketni manzilga olib boradigan yo'l ma'lumoti
- Router har paketni marshrut jadvali bo'yicha uzatadi
- Har qurilma qarorni mustaqil qabul qiladi — estafeta kabi

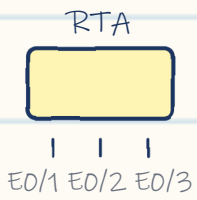


1-rasm. Paket hop-by-hop, optimal yo'l bo'ylab uzatiladi

Qoida: router paketni faqat keyingi qadamga beradi — undan keyin nima bo'lishi keyingi routerning ishi.

2. Marshrut jadvali

Har router o‘z jadvalini saqlaydi; uzatish faqat shu jadval bo‘yicha.



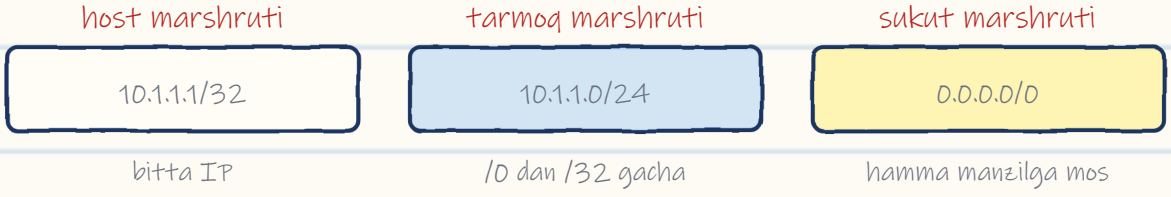
Manzil/Maska	Next hop	Port	Metrika
0.0.0.0/0	20.0.0.2	E0/2	10
10.0.0.0/24	10.0.0.1	E0/1	0
20.0.0.0/24	20.0.0.1	E0/2	0
40.0.0.0/24	20.0.0.2	E0/2	1
40.0.0.0/8	30.0.0.2	E0/3	3

qayerga · kim orqali · qaysi port · qancha xarajat

2-rasm. Bitta yozuv — bitta yo‘nalish uchun ko‘rsatma

- Manzil/maska — maqsad tarmoq (manzil VA maska)
- Next hop — keyingi routerning IP manzili
- Port — paket chiqadigan interfeys
- Metrika — shu yo‘lning xarajati

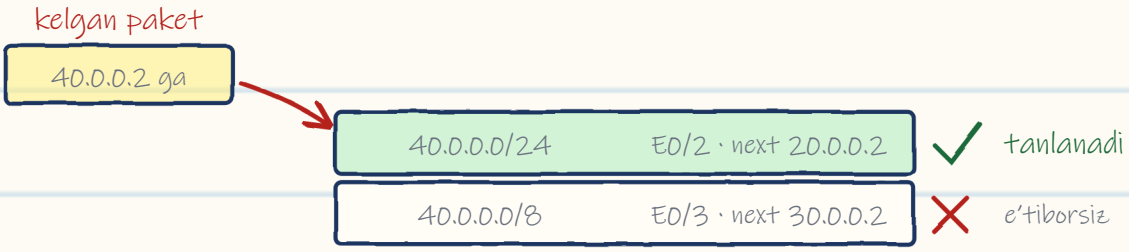
3. Maska uzunligi bo‘yicha turlari



maska uzunligi = aniqlik darajasi

3-rasm. Maska uzun bo‘lsa — yo‘l aniqroq

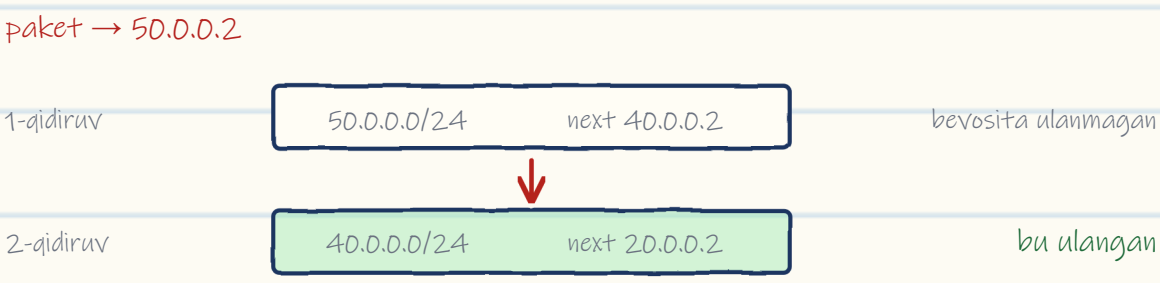
4. Yo‘l tanlash qoidalari



eng uzun maska yutadi (longest match)

ikkitasi ham mos, lekin /24 aniqroq

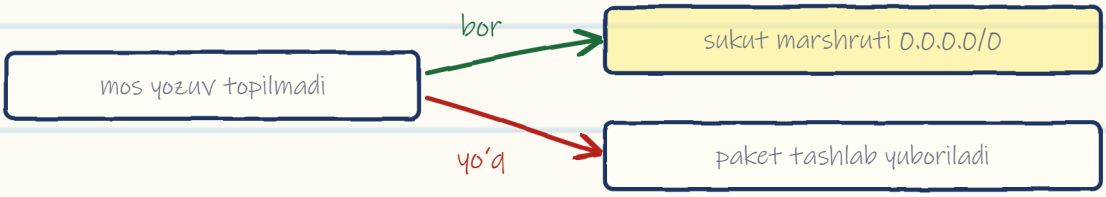
4-rasm. Bir necha mos yozuv bo‘lsa — maskasi uzuni ishlatiladi



rekursiv qidiruv — ulangan next hop topilguncha

so‘ng paket E0/2 orqali chiqadi

5-rasm. Next hop ulanmagan bo‘lsa — jadval yana qidiriladi

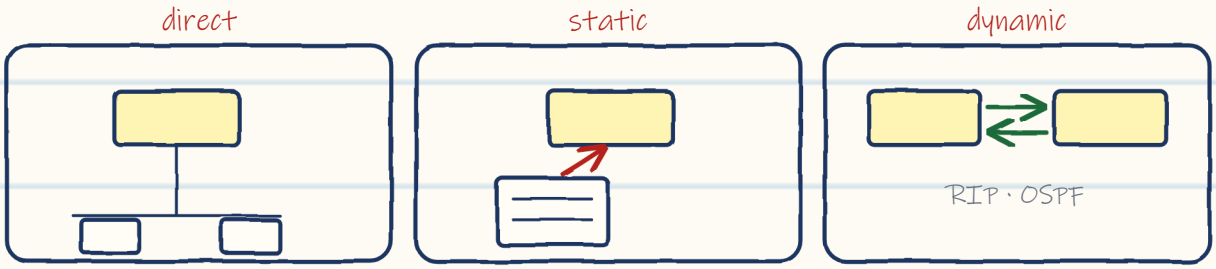


sukut marshruti — eng qisqa maska, oxirgi variant

0.0.0.0/0 hamma manzilga mos keladi

6-rasm. Hech narsa mos kelmasa — sukut marshruti yoki tashlash

5. Marshrut qayerdan keladi?

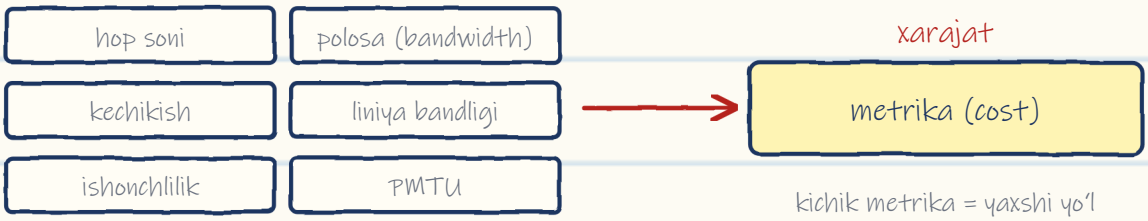


direct	static	dynamic
interfeys o'zi yasaydi	qo'lda yoziladi	protokol o'zi topadi
xarajat kichik	metrika = 0	resurs ko'p ketadi
faqat ulangan tarmoq	oddiy tarmoq uchun	murakkab tarmoq uchun

nosozlikda static yangilanmaydi, dynamic o'zi qayta hisoblaydi

7-rasm. Uchta manba: ulanish, qo'l va protokol

6. Metrika — yo'lning xarajati



kichik metrika = yaxshi yo'l

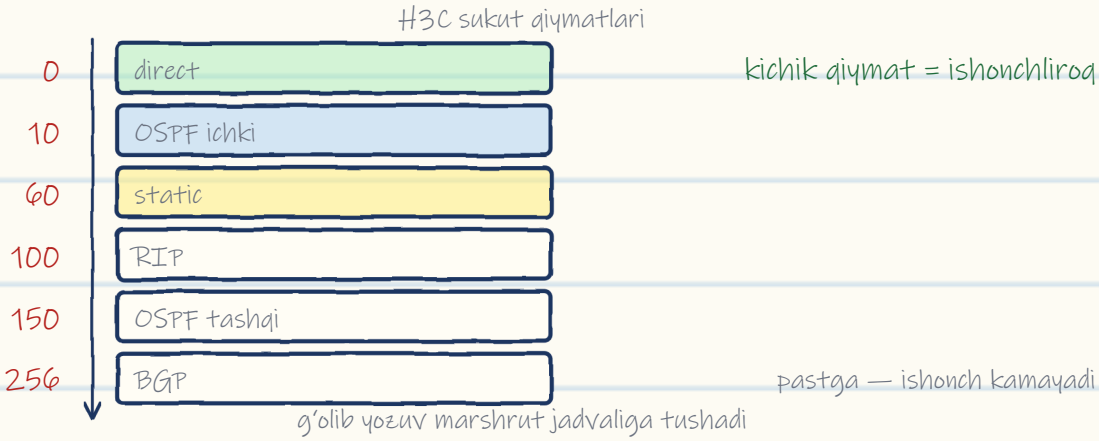
har protokol o'z metrikasini hisoblaydi — ular taqqoslanmaydi

8-rasm. Metrika bir necha omildan hisoblanadi

Protokol	Metrika asosi
static	doimiy 0
RIP	hop soni
OSPF	polosa (bandwidth)

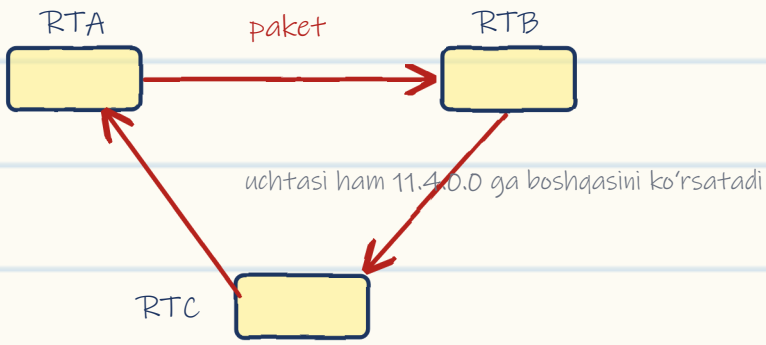
7. Protokol ustuvorligi (preference)

Bir manzilga bir necha protokol yo'l topsa — preference solishtiriladi.



9-rasm. Preference — protokolning ishonchlilik darajasi

8. Marshrut halqasi (loop)



sabab: noto'g'ri static yoki yomon reja

TTL = 0 bo'lguncha aylanadi — polosa behuda ketadi

10-rasm. Halqada paket manzilga yetmaydi, faqat aylanadi

9. Jadvalni ko'rish

Buyruq	Nima ko'rsatadi
display ip routing-table	qisqa jadval
display ip routing-table 1.1.1.1	aniq bir marshrut
display ip routing-table statistics	marshrut statistikasi

display ip routing-table

Destination/Mask	Proto	Pre	Cost	NextHop	Interface
0.0.0.0/0	Static	60	0	20.0.0.2	GE0/1
10.0.0.0/24	Direct	0	0	10.0.0.1	GE0/0
20.0.0.0/24	OSPF	10	2	20.0.0.1	GE0/1

Pre — protokol ustuvorligi, Cost — metrika

11-rasm. Chiqishda Proto · Pre · Cost ustunlariga qaraladi

Xulosa

- Marshrut yo'lni ko'rsatadi, router hop-by-hop uzatadi
- Yozuv: manzil/maska · next hop · port · metrika
- Tanlash: eng uzun maska → rekursiv qidiruv → sukut
- Manba: direct · static · dynamic (RIP, OSPF)
- Preference kichik = ishonchli; halqada TTL nolga tushadi

Tekshirish

1. Sukut marshruti maskasi? — 0.0.0.0/0
2. Ikki mos yozuvdan qaysi biri? — maskasi uzuni
3. Static marshrut preference va metrikasi? — 60 va 0
4. RIP va OSPF metrikasi nimadan? — hop soni, polosa